

# Matias Barandiaran

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(+49) 152 3162 6943 • [mbarandiaranr@gmail.com](mailto:mbarandiaranr@gmail.com) • [mbaranr.github.io](https://mbaranr.github.io)

## Education

### Lancaster University

#### Bachelor of Science in Computer Science

Leipzig, Germany

October 2022 – July 2025

- Ranked **1st** out of **212** students.
- First Class Honours (**UK**: 91.2%  $\approx$  **DE**: 1.0  $\approx$  **GPA**: 4.0/4.0).
- University ranked **#132** worldwide for Computer Science (QS by subject) in 2025.

### San Ignacio de Recalde School

#### International Baccalaureate

Lima, Peru

March 2020 – December 2022

- Cohort's top 5% (**GPA**: 3.9/4.0).

## Academic Gaps

**August 2025 – Present:** Presented at conference and enrolled in German language lessons.

**January 2022 – September 2022:** Transition period between end of Peruvian school year and start of UK undergraduate year. Completed visa processing, relocation, and academic preparation.

## Work Experience

### Technical University of Munich

#### Research Assistant (Hiwi)

Munich, Germany (Hybrid)

July 2024 – August 2025

- Hosted by the Gagneur Lab and Helmholtz Munich.
- Worked on rare variant association testing using deep learning and data-driven burden scores (DeepRVAT), utilizing **Python**, **R**, **PyTorch**, and **Snakemake**.

### Martin Luther University of Halle-Wittenberg

#### Research Intern

Remote

July 2024 – September 2024

- Hosted by the Faculty of Medicine.
- Worked on graph-level anomaly detection via latent space searching and generative adversarial networks (gladGAN) on molecular datasets, utilizing **Python**, **PyTorch**, and **scikit-learn**.

### Lancaster University Leipzig

#### Teaching Assistant

Leipzig, Germany

October 2023 – June 2024

- Delivered instructional sessions of **MIPS** assembly programming for the Digital Systems module.

### Charles University

#### Research Intern

Prague, Czechia

August 2023 – September 2023

- Hosted by the Faculty of Mathematics and Physics.
- Contributed to paper “Unsatisfiability Proofs for Horn Solving”. \*ETAPS best paper award\*
- Designed production of Alethe verifiable proofs for the GOLEM horn clause solver in **C++**.

## Freelance

Lima, Peru

## Math Tutor

August 2020 – September 2021

- Guided senior high school students to achieve outstanding exam performance and substantial growth in their mathematical abilities (IB Mathematics SL and HL).

## Publications

Barandiaran, M. & Stovold, J. (2025). Growing Reservoirs with Developmental Graph Cellular Automata. In Proceedings of International Conference on Artificial Life 2025 (pp. 26–35). MIT Press.

Otoni, R., Blicha, M., Barandiaran, M., Eugster, P., Kofrön, J., & Sharygina, N. (2025, May). Unsatisfiability Proofs for Horn Solving. In Proceedings of International Conference on Tools and Algorithms for the Construction and Analysis of Systems (pp. 67-87). Springer Nature.

## Awards / Honours

**Chancellor's Medal** – Lancaster University (2025): The university's most prestigious award, presented annually to graduating students who demonstrate exceptional merit in their field.

**Valedictorian** – Lancaster University (2025): Ranked **1st** out of **212** students achieving the cohort's highest average grade across all three years of the BSc Computer Science program.

**Outstanding Dissertation** – Lancaster University (2025): Obtained the highest achievable grade (**100%**) for the BSc Computer Science dissertation. **Strong Accept** for presentation at peer-reviewed International Conference on Artificial Life 2025 in Kyoto, Japan.

**EASST Best Paper Award** – Charles University (2025): Best paper related to the systematic and rigorous engineering of software and systems at peer-reviewed International Joint Conferences On Theory and Practice of Software 2025 in Hamilton, Canada.

**Scholarship** – Lancaster University (2022): Awarded a €4,000 tuition scholarship in recognition of academic merit.

## Leadership Activities

### Lancaster University

Leipzig, Germany

### Founder, Computer Science Society

2023 – 2025

- Dedicated to fostering a culture of innovation, diversity and problem-solving.
- Secured university funding for the purchase of robotics supplies (e.g., 3D printer, Arduinos, etc.).

### Lancaster University

Bailrigg, England

### Co-Organizer, Global Advancement Fund

2024

- Co-Initiated and championed the approval process of a ~€20k fund to facilitate international collaboration between Lancaster campuses worldwide.

### Lancaster University

Leipzig, Germany

### Lead Organizer, Qiskit Fall Fest

2023

- Quantum Computing (QC) event powered by IBM.
- Gave an introductory talk on QC and held workshops with Qiskit material. Assembled a series of seminars with special guests like Enrique Solano (Co-CEO of Kipu Quantum).

### **Certifications**

*AWS Certified AI Practitioner* – Amazon Web Services (2025): Demonstrated proficiency in applying foundational AI/ML concepts, model deployment workflows, and cloud-native AI services within AWS.

*PyTorch for Deep Learning* – DeepLearning.AI (2025): Hands-on training in building, fine-tuning, and deploying deep learning models in PyTorch across vision and NLP, covering modern architectures, optimization, interpretability, and production tools such as ONNX, MLflow, pruning, and quantization.

*Prompt Engineering for Developers* – DeepLearning.AI & OpenAI (2025): Covered techniques for designing effective prompts, controlling model behaviour, and building production-grade LLM applications through the OpenAI API.

*Advanced Applications of Neural Networks and Deep Learning* – University of Oxford (2023): Explored Generative Adversarial Networks, Variational Autoencoders, Deep Reinforcement Learning, Diffusion Models, and Graph Neural Networks.

*Artificial Intelligence and Machine Learning: Theory and Practice* – University of Oxford (2023): Studied the underlying mathematical foundations of linear and convolutional neural networks.

### **Additional Skills and Interests**

**Languages:** Spanish (native), English (C2) \*TOEFL: 114/120\*, German (A2.1).

**Tools:** Python, Java, R, PyTorch, Lightning, TorchVision, Optuna, Git, C, C++, SQL, Linux, Snakemake, QEMU, Amazon Bedrock, Polars.

**Concepts:** Agile Development, Continuous Integration, Data Pipelines, Computer Vision, Generative AI, Supervised Learning, Reinforcement Learning, Reservoir Computing, Cloud Computing, Retrieval-Augmented Generation, Natural Language Processing, Model Finetuning, Prompt Engineering.

**Hobbies:** Game Development, Fishwatching, Badminton.